

VLAN Firewall Matrix – Reference Build

Designing and Building the Modern Smart Home (HP Z440 + RTX 5060 Ti)

This matrix defines the intended firewall policy for the segmented network described in Chapter 3. Default policy is DENY all inter-VLAN. Only explicitly allowed traffic is permitted. OPNsense is stateful, so return traffic is automatic.

Test goal: Verify that each rule does what it says — can a device on IOT actually NOT reach TRUSTED, etc.

1. VLAN Definitions (Reference)

VLAN ID	Name	Subnet (Example)	Purpose	Example Clients
10	MGMT	10.10.10.0/24	Proxmox, OPNsense, Switches, APs, ZFS management	Proxmox host, managed switches
20	TRUSTED	10.10.20.0/24	User devices, Home Assistant, NAS	PCs, phones, Home Assistant VM
30	IOT	10.10.30.0/24	General IoT, cloud-capable devices	Tasmota plugs, bulbs, ESPHome
40	GUEST	10.10.40.0/24	Guest WiFi, isolated, internet only	Guest phones

VLAN ID	Name	Subnet (Example)	Purpose	Example Clients
50	CAM	10.10.50.0/24	IP Cameras, NO internet	Reolink, Amcrest cams
60	IOT_RESTRICTED	10.10.60.0/24	No-internet IoT, local only	Cheap sensors, vacuums

DNS for all VLANs: Redirect to OPNsense / AdGuard Home on 10.10.10.1
 NTP for all VLANs: Allow to OPNsense only

2. Core Principles

1. Guest and Cameras can NEVER initiate to other VLANs.
2. IOT can NEVER initiate to Trusted or MGMT.
3. Trusted can initiate to ALL (for management).
4. Cameras have NO internet access. Frigate pulls RTSP.
5. All VLANs DNS is forced to firewall, not 8.8.8.8.

3. Firewall Matrix

#	Source	Destination	Port / Proto	Action
MGMT (10)				
1	MGMT	ANY	ANY	ALLOW
TRUSTED (20)				

#	Source	Destination	Port / Proto	Action
2	TRUSTED	MGMT	8006 TCP, 22 TCP, 443 TCP, 80 TCP	ALLOW
3	TRUSTED	IOT	ANY	ALLOW
4	TRUSTED	CAM	554 TCP, 8000 TCP, 80 TCP	ALLOW
5	TRUSTED	GUEST	ANY	ALLOW
6	TRUSTED	Internet	ANY	ALLOW
IOT (30)				
7	IOT	TRUSTED	ALL	BLOCK
8	IOT	MGMT	ALL	BLOCK
9	IOT	CAM	ALL	BLOCK
10	IOT	IOT	ANY	ALLOW
11	IOT	TRUSTED - Home	8123 TCP,	ALLOW

#	Source	Destination	Port / Proto	Action
		Assistant IP	1883 TCP, 8883 TCP	
12	IOT	Internet	443 TCP, 80 TCP, 123 UDP	ALLOW
GUEST (40)				
13	GUEST	RFC1918 (10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16)	ALL	BLOCK
14	GUEST	Internet	ANY	ALLOW
CAM (50)				
15	CAM	ANY RFC1918	ALL	BLOCK
16	CAM	Internet	ALL	BLOCK
17	CAM exception	TRUSTED - Frigate NVR IP	554 TCP (if cam needs to push)	BLOCK in ref build
IOT_RESTRICTED (60)				
18	IOT_RESTRICTED	Internet	ALL	BLOCK
19	IOT_RESTRICTED	TRUSTED - HA IP	1883 TCP	ALLOW

#	Source	Destination	Port / Proto	Action
Common Services (All VLANs)				
20	ALL VLANs	OPNsense	53 TCP/UDP	ALLOW + NAT REDIRECT
21	ALL VLANs	OPNsense	123 UDP	ALLOW
22	IOT, CAM, IOT_RESTRICTED	OPNsense	67/68 UDP	ALLOW

4. OPNsense Implementation Notes

- Create aliases: RFC1918 , HA_IP (10.10.20.10) , Frigate_IP (10.10.20.30) , DNS_Server
- Rules are applied per interface (VLAN). The CAM interface should have final BLOCK all to RFC1918 + BLOCK all to Internet.
- For mDNS: Enable Avahi plugin on OPNsense and allow reflection between TRUSTED and IOT only, not GUEST or CAM.
- For Home Assistant discovery: Allow UDP 5353 mDNS from IOT to HA_IP.

5. How To Test (For Contributor)

Test from a device on each VLAN. Log results like this:

```
[TEST] 2026-09-29 | Source: IOT (10.10.30.50) > Dest: TRUSTED (10.10.20.10:8123) | Expected: ALLOW | Result: PASS | Notes: HA reachable
```

[TEST] 2026-09-29 | Source: IOT (10.10.30.50) > Dest: TRUSTED (10.10.20.5:445) | Expected: BLOCK | Result: FAIL | Notes: Could ping laptop, rule 7 not working

Checklist:

- ☐ IOT cannot ping TRUSTED PC
- ☐ IOT CAN ping Home Assistant on 8123 and 1883
- ☐ CAM cannot ping 8.8.8.8
- ☐ CAM cannot ping IOT or TRUSTED (Frigate should still get stream)
- ☐ GUEST cannot ping any 10.10.x.x
- ☐ TRUSTED can ping everything
- ☐ DNS leak test: set device DNS to 8.8.8.8 on IOT, does it still resolve via AdGuard? Should.

Please open a PR with results in `/tests/fw-matrix-results.md`

6. Acknowledgments

Contributors to this matrix will be credited in the next edition of the book.

Maintained by Alan Cuzen - Reference build HP Z440 + RTX 5060 Ti - Proxmox VE 8.x / OPNsense 24.x / Frigate 0.14 + TensorRT